eProject · Team preparation

Role Preparation Guide

Nexus Service Marketing System - what each member shows in the demo and must be able to explain in the viva

#	Member	Student ID	System role	Presents
1	**Ali Jan**	Student1661815	**Admin**	Slides 1-5: introduction, problem and solution
2	**Humaiz Khan**	Student1659420	**Retail Staff**	Slides 6-8: architecture, database and identifiers
3	**Muhammad Ehtisham Shakeel**	Student1657963	**Technical Staff**	Slide 9 and the live demo
4	**Muhammad Waleed**	Student1659932	**Accounts Staff**	Slides 10-12: billing, plans and postpaid control
5	**Ali Turab**	Student1667727	**Customer**	Slides 13-17: interface, quality and closing

How to use this guide

Read the **Everyone** section first - every member must know it.

Then study **your own role** section: practise the demo steps on the running system until you can do them without reading.

Ask each other the viva questions. If you cannot answer one, open the file listed under **Files to know** and read the comments - every block is commented.

ADSE (ACCP in AI - Term 3) · ACE-PK-Karachi-20-North Nazimabad-2 · EPROJECT00503 · Mentor: Sir Zafeer Ansari

Everyone - the whole project in 2 minutes

What the project is

Nexus Communications sells **Dial-Up**, **Broadband** and **Landline** connections through a retail shop in every city. All records were kept on paper. Our web application keeps every step online: **order → feasibility check → connection → monthly postpaid bill → payment**, with a separate secure login for each of the five roles.

Technology - say it like this

Part	What we used
Language / framework	C#, ASP.NET Core 8 MVC (Model - View - Controller)
Database	SQL Server with Entity Framework Core 8 (code-first, migrations); 19 tables
Front end	Razor views, Bootstrap 5, our own CSS and JavaScript (light / dark theme, animations)
Structure	One Area per role (Admin, Retail, Technical, Accounts, Customer); business rules in Services (BillingService, OrderWorkflowService, ConnectionProvisioningService)
Security	Cookie login, role-based authorization, passwords hashed with PBKDF2 (salted), anti-forgery tokens on every form

The five roles

Role	Main job
Admin	Master data: cities, retail shops, employees, plans, discount schemes, vendors, products, purchases; reports
Retail staff	Register walk-in customers, place orders, collect bill payments (own shop only)
Technical staff	Feasibility checks, create connections, equipment, suspend / reactivate overdue connections
Accounts staff	Generate postpaid bills, record payments made at the office
Customer	Register, place and track orders, see connections and bills, upload documents, feedback

Numbers everyone must know

Item	Rule
Order number	11 characters: service letter + 10 digits in serial order - D (Dial-Up), B (Broadband), T (Telephone). Example D0000000001
Account ID	16 characters: service letter + 3-digit city code + 12-digit serial. Example B001000000000001 (B, Karachi 001, connection 1)
Security deposit	Dial-Up $325, Broadband $500, Landline $250 - refundable when the connection is closed
Bulk discount	10-14 connections 25%, 15-24 50%, 25-49 75%, 50+ 100% - on the advance (plan fee) and the security deposit

Item	Rule
Service tax	12.24% on the whole bill
Connection status	Active, Temporarily inactive (suspended), Permanently inactive (closed)
Order status	Placed → Under feasibility check → Feasible / Not feasible → Connected (or Cancelled)

Live demo - one order through the whole system

1. **Humaiz Khan** (retail1) places a Broadband order for a customer and reads out the order number.
2. **Muhammad Ehtisham Shakeel** (tech1) starts the feasibility check, records Feasible and provisions the connection - the account ID appears.
3. **Muhammad Waleed** (acct1) generates the first bill and explains the charges preview.
4. **Humaiz Khan** (retail1) collects the payment - the bill becomes Paid.
5. **Ali Turab** (alice) shows the customer side: bills, documents, light / dark theme; then tracks the order on the public Track page.
6. **Ali Jan** (admin) closes with the dashboard and the Reports page.

> **If the live demo fails**
>
> Stay calm and say: "Let us show you the recorded demonstration." Then play Video Demonstration\NexusServiceMarketingSystem_Demo.mp4.

Questions anyone may be asked

Question	Answer
Why ASP.NET Core MVC?	It separates data (Models), screens (Views) and logic (Controllers), has built-in security (login, anti-forgery) and works directly with SQL Server through Entity Framework.
What is Entity Framework Core?	An ORM: we write C# classes (entities) and EF creates the tables through migrations and turns our LINQ queries into SQL.
How does each role see only its own pages?	Each role has its own Area, and its controllers only let that role in ([Authorize] attributes); customer pages also read only the logged-in customer's data. Anyone else gets Access denied.
How are passwords stored?	Never as plain text: they are hashed with PBKDF2 and a random salt (ASP.NET Core PasswordHasher).
How did you test it?	48 automated end-to-end checks drive every role through the real system on SQL Server, plus browser checks for phone width, reduced motion and the theme switch. All pass.
Is it responsive?	Yes - Bootstrap grid. On a phone the menu becomes a ☰ button, cards stack and tables scroll inside their box. Show it with F12 → Ctrl+Shift+M.

Admin - Ali Jan

Item	Details
Member	Ali Jan (Student1661815)
System role	**Admin**
Demo login	admin / Admin@123
Presentation	Slides 1-5: introduction, problem and solution
Development area	Requirement analysis, admin module and project coordination

What this role does

The Admin (manager) maintains all master data and watches the whole business. The specification says the employee, stock, vendor and retail-shop details are maintained by the Admin **solely**.

Demo steps

1. Log in as admin. Point out the live figures: customers, active connections, open orders, low-stock items.
2. **Master data → Cities**: each city has a 3-digit code (Karachi 001 …) that becomes part of every account ID.
3. **Master data → Employees → New employee**: choose the role (Accounts, Technical, Retail) and, for retail, the shop. This creates the staff login.
4. **Master data → Plans**: show add / edit / delete / search; every plan has price, validity, deposit and (landline) per-minute rates.
5. **Stock → Products**: stock quantity and reorder level; **Stock → Purchases**: recording a vendor purchase increases stock.
6. **Operations → Reports**: orders, connections, billing totals, equipment demand, low stock, customer feedback.

Key points to explain

- Only Admin can manage cities, shops, employees, plans, discount schemes, vendors, products and purchases.
- Low stock = stock at or below the product's reorder level - shown on the dashboard and in Reports.
- Equipment demand in Reports compares modems / routers needed by open internet orders with the stock, so Admin knows what to order (spec: order equipment based on requests received).
- Discount schemes hold the bulk bands (10-14 → 25% …); orders pick the band automatically.
- The Yearly STD landline price is blank in the specification, so that plan is stored inactive until Admin sets a price.
- Admin is created automatically on the first run (admin / Admin@123) - the password should be changed after the first login.

Files to know

File	What it does
Areas/Admin/Controllers/ EmployeesController.cs	Create / edit staff and their login
Areas/Admin/Controllers/ PlansController.cs	Plans add / edit / delete / search
Areas/Admin/Controllers/ PurchasesController.cs	Vendor purchases - adds to product stock
Areas/Admin/Controllers/ OperationsController.cs	Orders list, advanced search, reports, feedback
Data/Seed/RuntimeAdminSeeder.cs	Creates the first admin login

Viva questions

Question	Answer
Who can add employees?	Only the Admin, in Master data → Employees. The role decides which area the employee can open.
How does the system know stock is low?	Each product has a reorder level; when stock ≤ reorder level it is counted as low stock.
What happens when a purchase is recorded?	The purchase is saved and the product's stock quantity increases by the purchased quantity.
Why does a city need a code?	The 3-digit code is part of the 16-digit account ID, so the ID shows which city the connection is in.
Can a retail employee see another shop's data?	No - retail staff are linked to one shop and only see that shop's customers, orders and bills.

Retail Staff - Humaiz Khan

Item	Details
Member	Humaiz Khan (Student1659420)
System role	**Retail Staff**
Demo login	retail1 / Passw0rd!x
Presentation	Slides 6-8: architecture, database and identifiers
Development area	Database design, ER diagram, entities and migrations

What this role does

Retail staff work at the Nexus shop in their city. The specification says **only the retail outlet places orders** for walk-in customers, keeps track of their orders, connections, bills and the payments collected at the shop.

Demo steps

1. Log in as retail1 - dashboard shows this shop's orders, connections and bills to collect.
2. **Customers → New customer** (optional): register a walk-in customer.
3. **Orders → Place order**: choose customer, service (Broadband), plan, city, installation address and number of connections → **Place order**.
4. Read out the **order number** (for example B0000000002) - the customer can track it.
5. Later in the demo: **Collect payment** → choose the bill → enter the amount and method → **Record payment**. The bill becomes Paid.

Key points to explain

- Order number = service letter + 10 digits in serial order (D / B / T), made safely even when two shops order at the same moment.
- Dial-Up needs a Nexus landline: if the customer has none, a landline plan is ordered together (both are checked for feasibility).
- Bulk discount is chosen automatically from the number of connections the customer takes (10+ → 25% …).
- A payment is recorded against one bill and can never be more than what is still due on that bill.
- Shop isolation: retail staff see and collect only for their own shop's orders.

Files to know

File	What it does
Areas/Retail/Controllers/ OrdersController.cs	Place order and order list for the shop
Services/Workflows/ OrderWorkflowService.cs	Order rules: landline for dial-up, bulk discount, order number
Services/Identifiers/IdentifierGenerator.cs	Makes order numbers and account IDs in serial order
Areas/Retail/Controllers/ PaymentsController.cs	Collect payment and payment history

File	What it does
Database ER diagram (report, Figure 3)	Tables and relations - you present the database slides

Viva questions

Question	Answer
How is the order number generated?	A counter per service type in the IdentifierCounters table is increased inside a database transaction (MERGE with HOLDLOCK), so two orders never get the same number.
Why must Dial-Up have a landline?	The specification says a Nexus landline is compulsory for dial-up; the order carries a landline plan when the customer has no line.
How many tables are in the database?	19 related tables, created by EF Core migrations; a SQL script is included as well.
Can a customer pay more than the bill?	No - the system rejects a payment bigger than the amount still due on that bill.
What does the bulk discount reduce?	The advance (plan fee on the first bill) and the security deposit, by 25 / 50 / 75 / 100%.

Technical Staff - Muhammad Ehtisham Shakeel

Item	Details
Member	Muhammad Ehtisham Shakeel (Student1657963)
System role	**Technical Staff**
Demo login	tech1 / Passw0rd!x
Presentation	Slide 9 and the live demo
Development area	Login and security, orders, feasibility and connections

What this role does

Technical staff decide whether an address can be served (feasibility), create the connection, supply the equipment, and make connections inactive - temporarily (for unpaid bills) or permanently (closed).

Demo steps

1. Log in as tech1 → **Feasibility**: the new order is listed.
2. **Start checks** - the system creates the checks this order needs.
3. **Record result** → distance from the exchange, server availability, **Feasible**, remarks → **Save result**. The order becomes Feasible.
4. **Provision** → choose the modem / router and type the serial number (a phone number for a landline) → **Provision connections**.
5. Point at the new **16-character account ID** in Connections.
6. **Overdue**: connections with a bill past its due date - **Suspend**; when everything is paid they appear with **Reactivate**.

Key points to explain

- Which checks: Telephone → landline check; Broadband → internet check; Dial-Up → both when a new landline is ordered, internet only when the customer already has a Nexus landline.
- All checks feasible → order Feasible; any check not feasible → order Not feasible.
- Account ID = service letter + 3-digit city code + 12-digit serial, e.g. B005000000000001 (Broadband, Faisalabad 005, first one).
- Provisioning takes one modem / router per internet connection from stock; the deposit is stored with the bulk discount already applied.
- Plan change only to a plan of the same service type; replacing spoiled equipment adds a replacement charge to the next bill (billed once).
- Postpaid control: unpaid overdue bills → suspend (Temporarily inactive); all paid → reactivate.

Files to know

File	What it does
Areas/Technical/Controllers/ FeasibilityController.cs	Start checks, record results

File	What it does
Services/Workflows/ OrderWorkflowService.cs	StartFeasibilityAsync / CompleteFeasibilityAsync - which checks, order status
Services/Workflows/ ConnectionProvisioningService.cs	Create connections, account ID, stock, deposit, overdue / suspend
Areas/Technical/Controllers/ ConnectionsController.cs	Status change, plan change, replace equipment, overdue list

Viva questions

Question	Answer
What is checked in feasibility?	Distance from the exchange and, for internet, server availability - recorded for each landline / internet check.
When does an order become Not feasible?	As soon as any of its checks is recorded as Not feasible.
How is the account ID made?	Letter for the service + the city's 3-digit code + a 12-digit serial number per service and city.
What happens to stock when you provision?	One modem / router is taken from stock for each internet connection; you cannot provision without stock.
When is a connection suspended?	When it has a bill past its due date that is not fully paid; it is reactivated once all its bills are paid.

Accounts Staff - Muhammad Waleed

Item	Details
Member	Muhammad Waleed (Student1659932)
System role	**Accounts Staff**
Demo login	acct1 / Passw0rd!x
Presentation	Slides 10-12: billing, plans and postpaid control
Development area	Billing, payments, overdue control, tracking and search

What this role does

The accounts department calculates the charges and generates every postpaid bill - **only accounts can generate bills**. They also record payments made at the head office.

Demo steps

1. Log in as acct1 → **Generate bill**.
2. Choose the connection and the billing period start → **Show charges**.
3. Explain the preview: plan fee, call rates (landline), replacement charges, unpaid amount from earlier bills.
4. Enter period end, issue date, due date (and call minutes for a landline) → **Generate bill**.
5. **Bills & payments**: every bill with total, brought forward, paid, outstanding and a coloured status badge.

How a bill is calculated

Part	Rule
Plan fee	Charged once per validity period (a quarterly plan every 3 months, not on every bill)
Call charges	Landline only: minutes × the plan's rates (local, STD, messaging for mobiles)
Security deposit	Only on the first bill - already reduced by the bulk discount
Discount	Only on the first bill: plan fee × bulk discount %
Replacement charge	Spoiled equipment replaced by Technical - billed once, on the next bill
Sub-total	Plan fee + calls + other usage + deposit + replacements - discount
Service tax	12.24% of the sub-total
Total	Sub-total + tax
Brought forward	Shown for information: what is still unpaid on earlier bills (paid against those bills)

Key points to explain

- One bill per connection per period start; no bill for a permanently inactive connection.
- Bill status: Issued → Partially paid → Paid (payments can never exceed what is due).

- Connection status depends on the bill: overdue unpaid bills lead to suspension by Technical.

Files to know

File	What it does
Services/Workflows/BillingService.cs	All money rules: GenerateBillAsync, RecordPaymentAsync, preview
Areas/Accounts/Controllers/ BillsController.cs	Generate bill page, bills list, office payments
Models/Entities/Bill.cs	Every part of a bill saved separately (itemised)

Viva questions

Question	Answer
How is the tax calculated?	12.24% of the sub-total, then added to give the total.
Why is the plan fee not on every bill?	Plans have a validity (1, 3, 6 or 12 months); the fee is charged once and covers that whole period.
When is the deposit charged?	Only on the connection's first bill, with the bulk discount already applied.
What is brought forward?	The amount still unpaid on the connection's earlier bills, shown on the new bill for information.
Can anyone else generate a bill?	No - only the Accounts role can open the Generate bill page.

Customer - Ali Turab

Item	Details
Member	Ali Turab (Student1667727)
System role	**Customer**
Demo login	alice / Passw0rd!x
Presentation	Slides 13-17: interface, quality and closing
Development area	Interface design, customer documents, testing and documentation

What this role does

Customers use the self-service portal: register, place and track orders, see their connections, itemised bills and dues, upload documents and send feedback. You also present the interface - design, light / dark theme, responsiveness and testing.

Demo steps

1. Show the **Register** page (name, type, email, phone, address, city, username, password).
2. Log in as alice → dashboard: active connections, orders in progress, amount due, total orders.
3. **My bills**: itemised - plan, calls, deposit, discount, tax, paid and due.
4. **Documents**: upload a proof (PDF / PNG / JPEG up to 5 MB) with type and year; it is filed by year and city.
5. Click the **sun / moon** button - the whole site switches between light and dark, and stays that way on the next page.
6. Log out and open **Track**: enter the order number → its stage, feasibility and connection.
7. Optional: F12 → Ctrl+Shift+M to show the phone layout.

Key points to explain

- A customer can cancel an order until the connection is created (Placed, Under feasibility check or Feasible).
- Documents are stored outside the public website folder; only the customer, Admin and their retail shop can open them; the file type is checked by its content, not only by its name.
- Light / dark theme: saved in the browser (localStorage) and applied before the page appears, so it never flashes.
- Ten original illustrations re-colour themselves for each theme through CSS variables.
- Accessibility: keyboard focus, 4.5:1 text contrast, all animation off for users who choose reduced motion.

Files to know

File	What it does
Areas/Customer/Controllers/ OrdersController.cs	Place, track and cancel own orders

File	What it does
Areas/Customer/Controllers/ DocumentsController.cs	Upload / download documents by year and city
Views/Shared/_Layout.cshtml	Menu, theme switch button, footer - shared by every page
wwwroot/css/site.css	The design system: colours, light / dark theme, responsive rules
wwwroot/js/site.js	Animations, loader, theme switch, particle background

Viva questions

Question	Answer
How does the site remember light or dark?	The choice is saved in the browser's localStorage; a small script in the page head applies it before the page is drawn.
Is the website responsive?	Yes - Bootstrap 5 grid and our CSS; tested at 390, 768, 1024 and 1440 px with no sideways scrolling.
Can a customer see another customer's bill?	No - every customer page reads only the data of the logged-in customer.
What can be uploaded?	PDF, PNG or JPEG up to 5 MB: address proof, application form, payment receipt or other.
How did you test the system?	48 automated end-to-end checks on SQL Server plus browser checks; all pass. The checklist is in chapter 11 of the report.